To Renounce Liberty? A Public-Goods Experiment on (Un)Democratic Reversal

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Abstract

Democratic backsliding often proceeds through lawful choices that weaken constraints on later decisions. Yet we know far more about why citizens surrender institutional safeguards than about whether they reclaim them once the crisis that justified concentrated authority recedes. We studied this reverse movement in an incentivized public-goods experiment with 300 participants in Finland. Citizens repeatedly decided whether an unknown leader's proposal required group approval or took effect automatically. Automatic implementation initially offered higher productivity and assured coordination, making weaker constraints materially attractive. We then randomly assigned matching pools to continued crisis or structural recovery, which made approval equally productive while leaving the leader's authority unchanged. The central outcome was whether citizens who had authorized decisions without approval subsequently voted to restore it. The design treated democratic reversal as a consequential choice over decision rights rather than a declaration of democratic sentiment. It therefore identified whether structural recovery caused citizens to withdraw the support that sustains executive aggrandizement and whether democratically authorized losses of decision rights remained reversible.

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1 Research Question and Contribution

Democratic backsliding depends not only on leaders who weaken constraints but also on citizens who authorize those changes. One common form is *executive aggrandizement*, in which formally lawful decisions gradually reduce constraints on one-person authority. This study examined its demand-side microfoundation by asking whether citizens who supported surrendering an approval right later voted to restore it after a crisis receded. We define that within-person change in institutional support as **(*un*)democratic reversal**.

We build on the democratic backsliding and public-good games literatures. Contemporary backsliding often advances without ending elections because citizens may support elected officials or lawful rule changes that reduce the approval, oversight, or veto constraints governing later decisions (Bermeo 2016). The politically relevant step is therefore not necessarily an explicit vote against democracy. It can instead be a *democratic* vote that makes subsequent authority *less* constrained. Threat and policy benefits can increase tolerance for weaker constraints (Hetherington and Suhay 2011; Goodman 2021; Saikkonen and Christensen 2023). Laboratory evidence similarly shows that gridlock can increase support for executive special powers (Forteza, Mussio, and Pereyra 2024).

This literature mainly explains movement toward less constrained authority. We followed the reverse movement by observing whether the same citizens voted to restore an approval right after its crisis-based benefit receded. That second choice turned reversal into a consequential decision over an institutional constraint rather than a stated evaluation of democracy.

Public-good games supply an economic reason for weakening approval. Voluntary provision is vulnerable to free riding, which means keeping one's points while others fund the shared service. This individual incentive can erode cooperation, whereas centralized enforcement can coordinate contributions (Fehr and Gächter 2000; Fischbacher and Gächter 2010; Hamman, Weber, and Woon 2011). Experiments also show that voting over institutions changes their performance (Dal Bó, Foster, and Putterman 2010; Sutter, Haigner, and Kocher 2010; Markussen, Putterman, and Tyran 2014). We joined these literatures by making citizen approval both institutionally meaningful and economically costly.

In each round, five citizens began with 20 points each and voted between two rules for managing a shared public-services fund. Under both rules, the software selected one citizen uniformly at random after the vote. Each citizen therefore had the same one-in-five chance of leading that round. The leader proposed how many points every citizen placed in the fund and whether any collected points moved to the leader's payoff. The game multiplied the remaining fund points and divided the group return equally.

Under the approval-constrained rule (D), at least three of the other four citizens had to approve the complete proposal. Rejection returned the group to voluntary individual allocations. Under automatic implementation (A), the same proposal took effect without that approval vote. Interaction occurred inside a stable *matching pool* of 5, 10, or 15 citizens. Larger pools were reshuffled into temporary groups of five before each round. A five-person pool played together throughout. The institutional ballot therefore selected an approval rule, not a known leader or proposal.

Everyone began with a public-service crisis. Participants read that "*Parliament has failed to pass a budget for eight months, healthcare waiting times have doubled, and planning for public services has stalled.*" During Block 1, each fund point created 1.50 group points under D and 2.50 under A. Direct implementation therefore had a one-point *productivity premium* and guaranteed that the leader's proposal was implemented. After Block 1, each matching pool received **institutional recovery** or **institutional persistence**. Recovery combined improved public-service conditions with a rise in the D rate to 2.50. Persistence retained the crisis and the 1.50 rate. The A rate and decision rule remained fixed.

The payoffs made the choice consequential. **The *allocation of decision rights* connected it to democratic backsliding**. A citizen who voted for A supported surrendering an approval vote before knowing the leader or proposal. A later vote for D supported restoring that safeguard after the crisis-based productivity advantage disappeared. *The experiment therefore observed whether citizens voted to take back a specific constraint on authority rather than inferring reversal from a survey declaration.*

In simple terms

Five citizens voted before seeing who would lead or what the leader would propose. Under D, three of the other four citizens had to approve the proposal. Rejection returned everyone to individual fund allocations. Under A, the proposal took effect directly. In Block 1, these methods used rates of 1.50 and 2.50, respectively.
After Block 1, recovery improved public-service conditions and raised D to 2.50. Persistence left the crisis and rate gap in place. Among previous A supporters, a later D vote supported restoring the approval right. Comparing recovery with persistence identified how much the structural recovery package changed that reversal.

Central insight

A decision can be democratically authorized while weakening the democratic constraints that govern what happens next. This resembles contemporary backsliding, which often proceeds through elections and formally lawful institutional changes rather than an outright abolition of democracy (Bermeo 2016).
In the experiment, the first majority vote was democratic. A citizen who chose A voted to let an unknown leader's later allocation and transfer take effect without the other citizens' approval. A majority for A removed that constraint while leaving the institutional ballot itself intact. A later vote for D supported restoring the approval right that the citizen had previously voted to surrender.

What matters is that citizens used one democratic decision to weaken their control over a subsequent decision and could later vote to take that control back.

2 Theory

2.1 Crisis and the productivity premium

A crisis can make action without approval attractive when ordinary procedures produce delay or failed coordination. The experiment gave this argument material consequences. In Block 1, proposals requiring approval (D) used a 1.50 fund rate, while proposals implemented directly (A) used 2.50. Under D, rejection also returned the group to voluntary contributions and possible free riding. Direct implementation therefore offered both a productivity premium and assured implementation of the leader's equal allocation (Fehr and Gächter 2000; Fischbacher and Gächter 2010; Hamman, Weber, and Woon 2011).

The personal-transfer option made the institutional constraint consequential. The leader could propose moving up to 20 fund points to their own payoff under either procedure. Under D, the other citizens saw that transfer before implementation and could reject it. Under A, they observed it only after it had taken effect. The experiment therefore concerned approval over discretion, not

the prevalence of corruption in Finland.

H1: Authorization under crisis. During Block 1, better outcomes under direct implementation (A) are associated with greater subsequent support, while larger observed transfers are associated with lower support. These are descriptive expectations about experience within the crisis environment.

2.2 Citizen authorization and democratic constraint

The institutional ballot occurred before leader selection and before the proposal. A vote for A therefore supported a rule rather than a known person or action. That ballot supported relinquishing participation in approving the next binding decision, whereas a vote for D supported retaining that right. **The design thus provided an abstract microfoundation of *executive aggrandizement* in which a *democratic* choice *weakened* the constraint governing what followed** (Bermeo 2016; Forteza, Mussio, and Pereyra 2024).

H2: Approval and discretion. Approval declines as proposed personal transfers increase. When leaders expect citizens to reject transfers despite the earnings from an approved proposal, leaders under D propose smaller transfers than those under direct implementation. These are behavioral expectations within endogenously chosen rules, not a separate randomized effect.

2.3 Recovery and democratic reversal

Recovery was assigned at the matching-pool level. Pool g contained 5, 10, or 15 participants. Five-person pools played together throughout, while larger pools were reshuffled into groups of five before each round. Three votes produced a majority within each group. The entire pool received the same Block-2 condition. We use $T_g = 1$ for recovery and $T_g = 0$ for persistence. Let m_{gb} denote the approval-constrained fund multiplier in pool g and block b. The productivity premium for direct implementation is

$$\delta_{gb} = 2.50 - m_{gb} \tag{1}$$

Everyone faced $\delta_{g1} = 2.50 - 1.50 = 1$ in Block 1. Recovery was implemented as one structural package. Participants learned that public-service conditions had improved, and m_{g2} rose to 2.50, which set $\delta_{g2} = 0$. Persistence reported that the crisis continued and kept $m_{g2} = 1.50$, so $\delta_{g2} = 1$. The leader-selection rule, proposal set, approval threshold, rejection-to-voluntary-allocation rule, and A rate remained fixed.

The recovery–persistence comparison identified the total causal effect of structural recovery on (un)democratic reversal. Among final-round Block-1 supporters of A, the analysis compared later D votes in recovery pools ($T_g = 1, \delta_{g2} = 0$) with those in persistence pools ($T_g = 0, \delta_{g2} = 1$). The public-service description stated what changed, and the production rate made that change materially consequential inside the game. Together, they defined the recovery package. The estimand therefore captured their combined effect rather than separate framing and payoff effects. Persistence supplied the counterfactual in which both the crisis and premium remained.

H3: Immediate restoration. Among participants who vote for A in Block 1's final round, recovery increases the probability of voting for D in Block 2's first round.

H4: Short-run reversal. Among the same participants, recovery increases the probability of voting for D in Block 2's fifth and final round, after four rounds of experience in the assigned environment.

3 Experimental Design

3.1 Overview

Each laboratory cohort completed one session of approximately 60 minutes with ten crisis rounds in Block 1 followed by five rounds in Block 2. The University of Turku issued a favorable ethical-review statement on 30 September 2026 (case TY/1722/2026).[1] Sessions took place at the UTU Choice Lab decision-making laboratory, which has 24 regular seats and one accessible seat. Production rosters contained 5, 10, 15, or 20 participants. Five-person sessions were an attendance fallback. Larger sessions retained anonymous rematching within stable ten- or fifteen-person pools. After Block 1, each entire pool was assigned to recovery or persistence. The pool then learned whether public-service conditions had improved or remained strained and saw its Block-2 fund rates. In the figures, p_i identifies participant i.

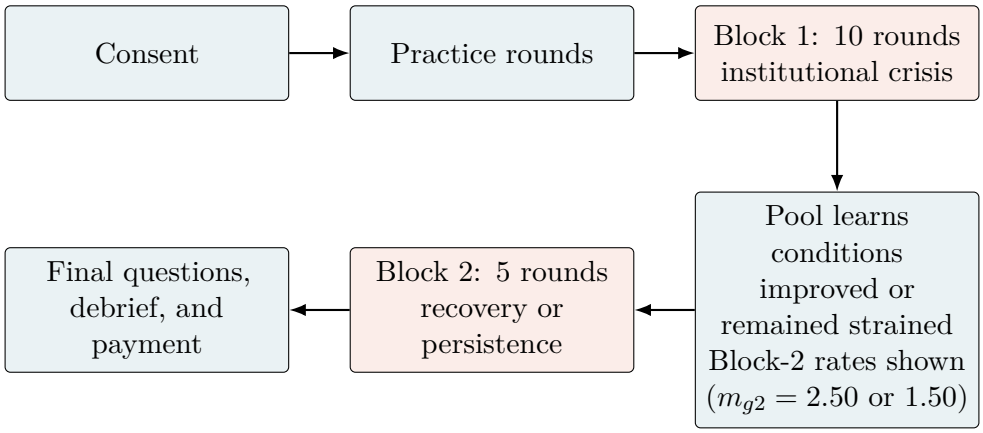


Figure 1: One-session sequence. Each paid round included an institutional ballot, leader proposal, implementation, and feedback. Larger pools were rematched between rounds.

[1] Protocol amendments require the applicable ethics review before implementation.

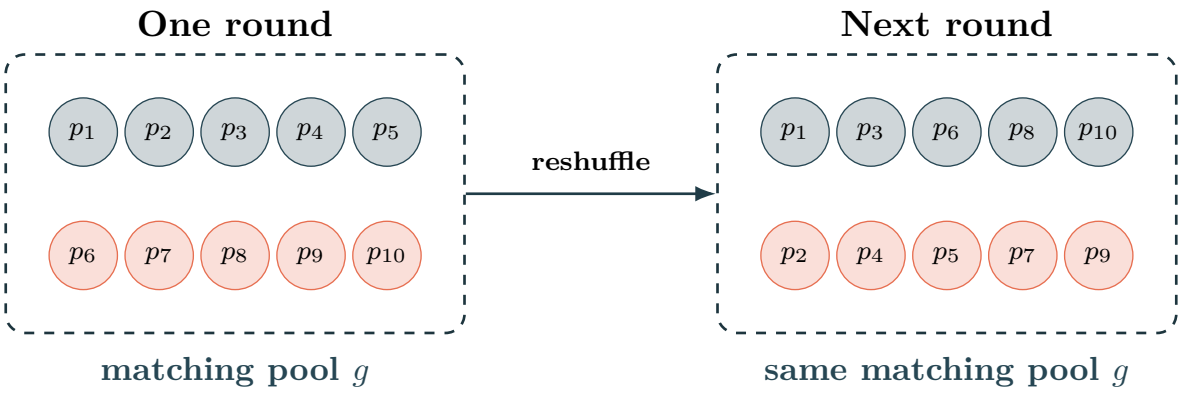


Figure 2: Anonymous rematching within a matching pool (ten-person example).
Notes: Each participant p_i remained in pool g. Ten- and fifteen-person pools formed two or three temporary groups per round. A five-person pool retained the same members throughout. Recovery or persistence applied to the whole pool in Block 2.

In simple terms

Ten- and fifteen-person pools permitted reshuffling within one treatment environment, limiting stable-team effects. When only five participants could start, they remained together throughout. Larger cohorts retained rematching rather than being split into fixed five-person teams. Reversal was measured for individuals, but treatment (T_g) was assigned to pools. Statistical uncertainty was therefore clustered by pool. Pool size was recorded because a fixed five-person group could develop a stronger shared history than a rematching pool.

Table 1: Session formation from the completed laboratory roster

People who arrive	Matching pools	Five-person groups per round	Participate	Released
20	$10 + 10$	$2 + 2 = 4$	20	0
16–19	15	3	15	1–4
15	15	3	15	0
11–14	10	2	10	1–4
10	10	2	10	0
6–9	5	1	5	1–4
5	5	1	5	0
Fewer than 5	—	—	0	All

Notes: Ten- and fifteen-person pools formed two or three temporary five-person groups per round. At 20 participants, two ten-person pools each formed two groups. Five-person pools retained the same membership. The roster was fixed before paid play, with no subsequent entry or replacement.

3.2 The institutional-choice public-good game

Each citizen started with 20 points and privately voted between an approval-constrained rule (D) and automatic implementation (A). Let $V_{i,b,r} \in \{\mathsf{D}, \mathsf{A}\}$ denote citizen i's ballot in block b and round r. Three votes selected the binding rule. The software then selected each group's leader through an independent uniform draw among its five citizens. Selection occurred under both rules and only after the institutional ballot. Every citizen therefore had the same chance before the draw in every round, and past service did not predict the next draw.

Table 2: Implemented design parameters

Feature	Implemented value
Laboratory session	One synchronized visit, approximately 60 minutes
Paid rounds	10 in Block 1 and 5 in Block 2 (15 total), preceded by unpaid practice
Group and matching pool	5-person groups within stable 5-, 10-, or 15-person pools
Rematching	Before each paid round in larger pools, with fixed membership in five-person pools
Round endowment	20 points per participant
Approval-constrained multiplier	1.50 in Block 1 and 2.50 under recovery or 1.50 under persistence in Block 2
Automatic-implementation multiplier	2.50 in both blocks and conditions
Approval	At least 3 of the 4 nonleaders, with a 2–2 split rejecting the proposal
Rejection	Voluntary individual contributions at the approval-constrained rate
Personal transfer	0–20 fund points, limited by the points collected
Performance bonus	Points earned in one randomly selected round from each block
Maximum performance bonus	EUR 5.00
Show-up payment	EUR 10.00
Maximum total payment	EUR 15.00

Selected-round points are converted linearly at EUR 1/24 per point. A round is selected uniformly within each block, so payment chances are 1/10 in Block 1 and 1/5 in Block 2. A selected round can yield at most 60 points, making the two-round bonus at most EUR 5. Both blocks retain equal weight in payment.

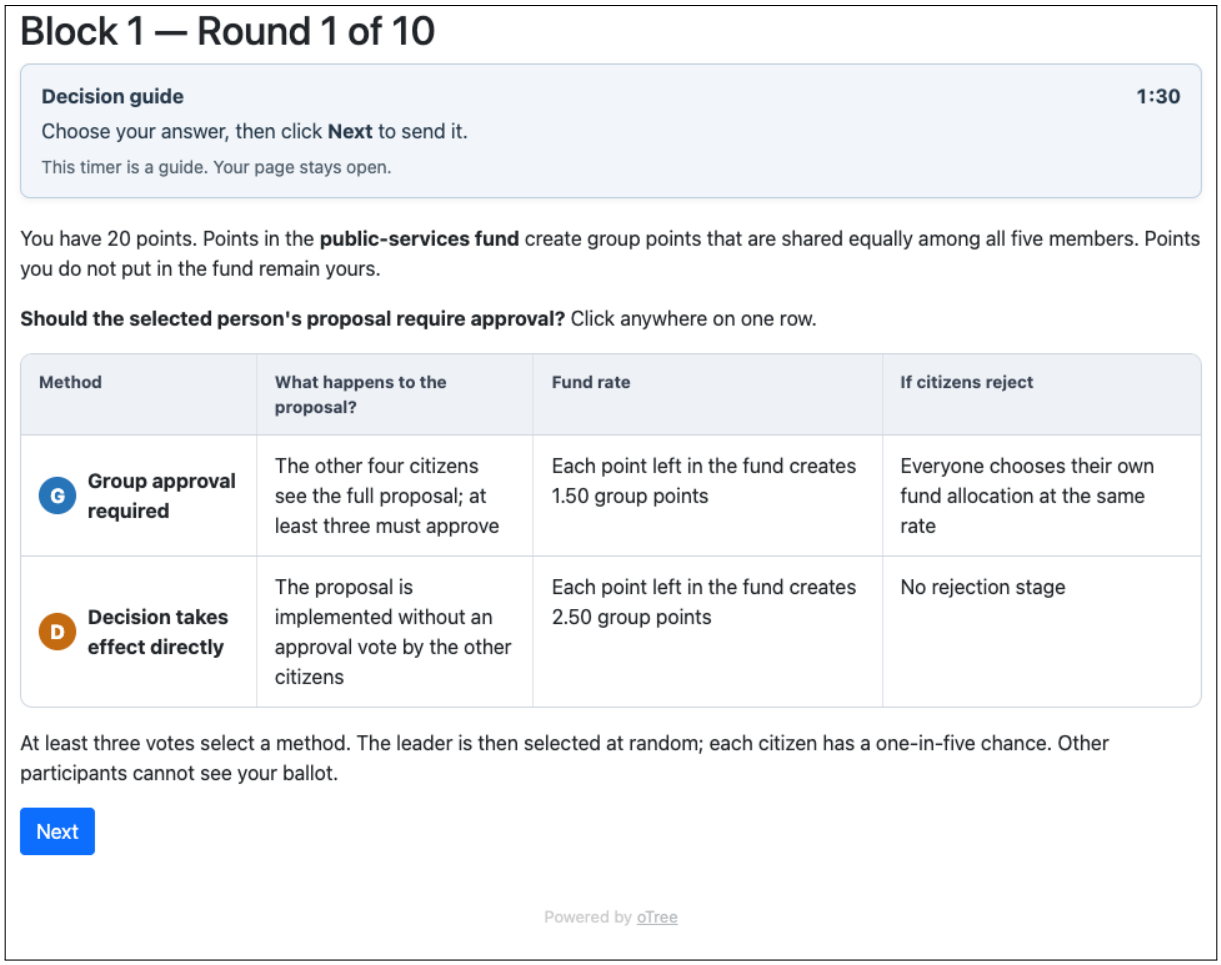


Figure 3: Participant screen for choosing whether the leader's proposal requires approval.

Leader proposal. Under either rule, the leader proposed an equal allocation $t \in \{0, \ldots, 20\}$ and a personal transfer $\rho \in \{0, \ldots, 20\}$, with $\rho \leq 5t$. Let π denote a citizen's payoff in that round. Under A, the proposal took effect directly, and a nonleader earned

$$\pi^{A}_{igbr} = 20 - t + \frac{2.50}{5}(5t - \rho), \tag{2}$$

and the leader earned $\pi^{A,L}_{igbr} = \pi^{A}_{igbr} + \rho$.

Approval-constrained procedure (D). The four nonleaders observed (t, ρ) and voted privately. Let $q_{igbr} = 1$ denote approval. The proposal passed when at least three approved. If it passed, the payoffs above applied with m_{gb} replacing 2.50. If it failed, each citizen chose $c_{igbr} \in \{0, \ldots, 20\}$ and earned

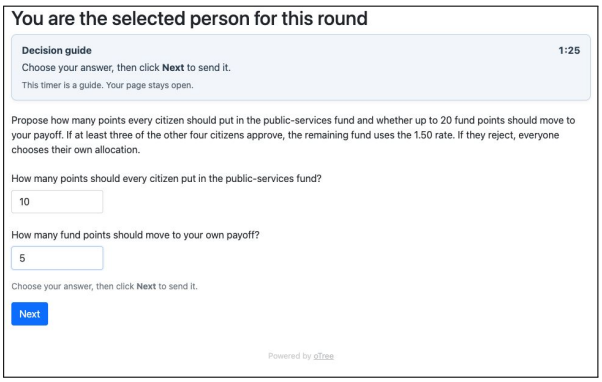
$$\pi^{D}_{igbr} = 20 - c_{igbr} + \frac{m_{gb}}{5} \sum_{j=1}^{5} c_{jgbr}. \tag{3}$$

The rejected transfer was not implemented. Everyone then saw the proposal, approval total, implemented allocation, transfer, and own payoff.

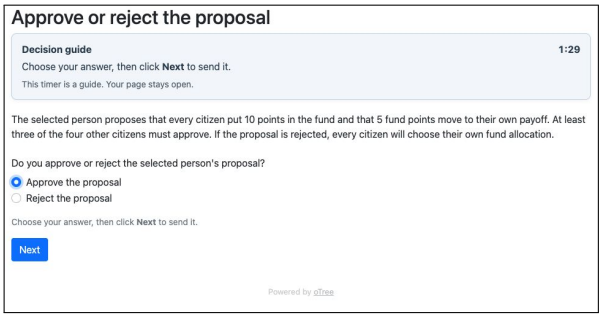
In simple terms

Rejection restored the standard **public-goods dilemma**. If one citizen contributed 1 point and four contributed 0, the contributor earned $19 + \frac{1.50}{5} = 19.30$ during the crisis, while each free-rider earned 20.30. Under recovery, these payoffs became 19.50 and 20.50. Contributing helped the group but cost the contributor. Direct implementation avoided that risk, whereas approval allowed citizens to block an unwanted proposal.

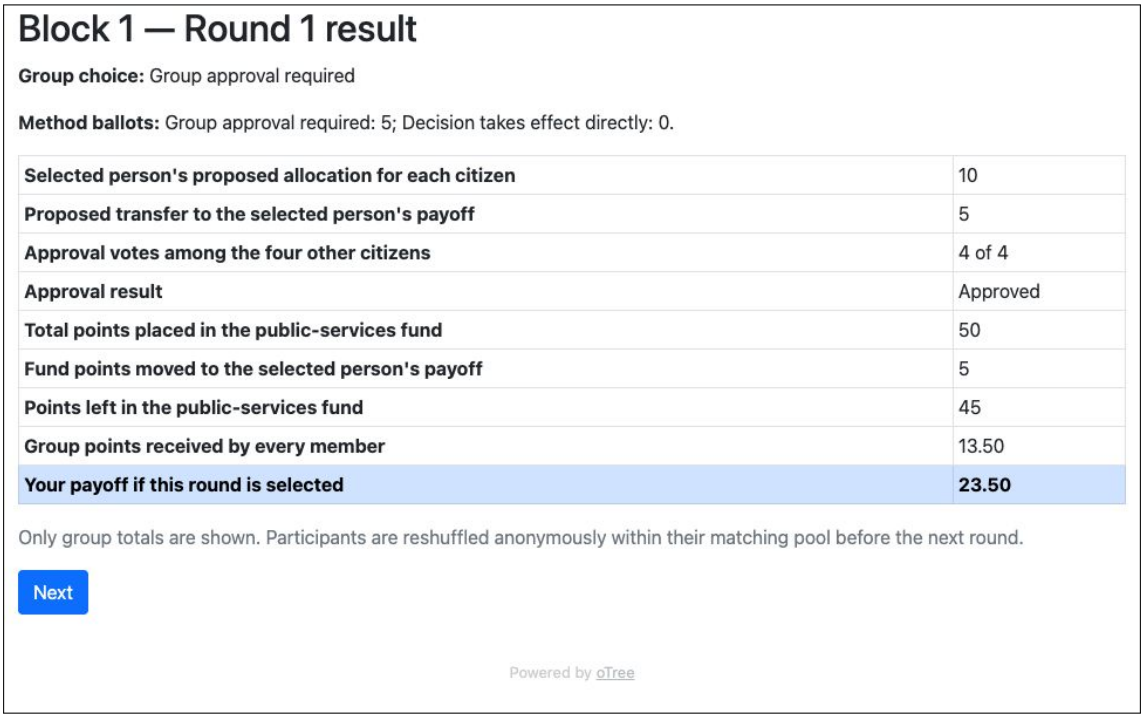
(a) Leader proposal



(b) Citizen approval



(c) Proposal, approval, and payoff feedback



Block 1 — Round 1 result

Group choice: Group approval required

Method ballots: Group approval required: 5; Decision takes effect directly: 0.

Selected person's proposed allocation for each citizen	10
Proposed transfer to the selected person's payoff	5
Approval votes among the four other citizens	4 of 4
Approval result	Approved
Total points placed in the public-services fund	50
Fund points moved to the selected person's payoff	5
Points left in the public-services fund	45
Group points received by every member	13.50
Your payoff if this round is selected	**23.50**

Only group totals are shown. Participants are reshuffled anonymously within their matching pool before the next round.

Next

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Figure 4: Proposal, approval, and round feedback. Entered amounts are illustrative.

How incentives map onto reversal. During Block 1, A combined a 2.50 rate with guaranteed implementation. By contrast, D used 1.50 and could return to voluntary contributions after rejection. Recovery reported improved public-service conditions and raised D to 2.50, while persistence retained

the crisis and rate gap. The proposal set, approval rule, rejection procedure, and A rule remained unchanged. The recovery–persistence contrast therefore estimated the total effect of this structural package.

From institutional choice to (un)democratic reversal. A citizen who moved from A to D first voted to let an unknown leader's proposal take effect without approval and later voted to restore that approval. The public-good incentives made this act consequential. After recovery, A still guaranteed implementation of the leader's equal allocation, whereas D could produce rejection and free riding. A higher switching rate under recovery would therefore indicate that structural change caused citizens to withdraw support for a weakly constrained decision rule. *Reversal was observed as a vote to restore a decision right rather than as a declaration of democratic sentiment.*

A simple payoff comparison

Suppose the leader proposed $t = 10$ and $\rho = 0$, and the proposal was approved under D. Holding that decision fixed gave the following payoffs.

	Approval required (D)	Direct implementation (A)
Crisis or persistence	25 points each	35 points each
Recovery	35 points each	35 points each

Recovery made an approved D proposal equally productive, but it did not make that proposal pay more. A still guaranteed implementation. **The question was whether citizens voted to restore approval once it became equally productive.**

Monetary benchmark. With own earnings as the sole objective, a leader maximizes the round payoff at $t = 20$ and $\rho = 20$. If citizens approve whenever acceptance pays more than rejection followed by free riding, this proposal passes under D. During the crisis, it pays the leader 44 and each nonleader 24, compared with 20 each after rejection. Under A, it pays 60 and 40, respectively. Recovery makes the approved D proposal identical to A, so both give an expected payoff of 44 before the random leader draw. The benchmark therefore favors A during the crisis and leaves the institutional vote tied after recovery. Lower transfers and rejected profitable proposals require behavioral explanations. Because approval requires three votes, nonpivotal ballots can also support weak voting equilibria. This calculation is a payoff-responsive benchmark rather than a unique equilibrium prediction.

Participants reported expected payoffs and transfers before the final Block-1 ballot and the first and final Block-2 ballots. Expected payoffs assessed whether they understood the rules' relative performance. Expected transfers traced beliefs about how leaders would use their discretion.

In simple terms

Five citizens cast private institutional ballots ($V_{i,b,r}$), after which the software selected a leader. That leader proposed (t, ρ). Under D, three of four nonleaders had to approve. Rejection activated individual contributions (c_{igbr}). Under A, the proposal took effect directly. Everyone saw the result and payoff before the next round. Only larger pools were rematched.

3.3 Block 1: crisis and leader authorization

After consent, participants practiced proposing, approving, rejecting, and making individual allocations after rejection. They then answered five comprehension questions. The interface never called either method democratic or undemocratic.

Block 1 began with the public-service crisis described above. Each fund point produced 1.50 under D and 2.50 under A. The leader could propose the same allocation and transfer under either rule. The rules differed in whether the other citizens could reject that proposal.

Each paid round repeated the same sequence. Citizens cast an institutional ballot, the leader made a proposal, and the proposal either faced approval or took effect directly. Rejection activated individual allocations before feedback and, in larger pools, rematching. We elicited expectations before the final ballot and selected one Block-1 round for payment.

3.4 Block 2: structural change and reversal

After Block 1, pools were assigned to *recovery* or *persistence*. Recovery reported that the budget had passed, waiting times were falling, and planning had resumed. It also raised the D multiplier from 1.50 to 2.50. Persistence retained the crisis and the 1.50 rate. The A rate, leader selection, proposal set, approval threshold, and rejection procedure remained unchanged. Together, the updated description and rate formed the structural-recovery treatment.

The first Block-2 ballot measured immediate support for restoring approval. The fifth and final ballot measured reversal after four rounds of experience and served as the primary outcome. We elicited expectations before both ballots and asked about democratic constraints only after paid decisions. One Block-2 round was selected for payment.

4 Outcomes, Process Measures, and Estimands

An *estimand* is the population-level quantity that an analysis seeks to recover. We first define the individual choices that compose the primary estimand.

Let $\mathbb{1}\{\cdot\}$ equal 1 when the condition inside braces is true and 0 otherwise. The final Block-1 ballot defines endorsement of automatic implementation as

$$A_{i1} = \mathbb{1}\{V_{i,1,10} = \mathsf{A}\}. \tag{4}$$

For the theoretically relevant subgroup with $A_{i1} = 1$, reversal is

$$R_i = \mathbb{1}\{V_{i,2,5} = \mathsf{D}\}. \tag{5}$$

The primary estimand is the average treatment effect of structural recovery among Block-1 supporters of automatic implementation:

$$\tau_R = \mathbb{E}[R_i(1) - R_i(0) \mid A_{i1} = 1], \tag{6}$$

where $R_i(1)$ and $R_i(0)$ are participant i's potential outcomes under recovery and persistence. Recovery includes both improved public-service conditions and the 2.50 D rate, so τ_R is the total effect of that package. Because A_{i1} was measured before assignment, subgroup membership was fixed before treatment. The primary randomization test evaluated the sharp null of no recovery effect for any eligible citizen by reassigning treatment according to the prespecified pool-level procedure. Because τ_R describes the average eligible citizen, the primary estimate weighted eligible participants equally. A secondary analysis gave each pool equal weight. Regression estimates

clustered uncertainty by pool and included prespecified indicators for five- and fifteen-person pools. Missing final ballots were reported by condition, with sensitivity checks retaining all randomized eligible citizens in the accounting.

Immediate reversal is

$$R_i^I = \mathbb{1}\{V_{i,2,1} = \mathsf{D}\} \quad \text{for } A_{i1} = 1. \tag{7}$$

The same comparison applied to R_i^I estimated immediate support for restoration. The final-round outcome R_i remained primary because it recorded support after further interaction. Choosing D at both endpoints indicated short-run persistence, without requiring an uninterrupted sequence of D votes.

In simple terms

The primary estimand (τ_R) asks how much recovery changed reversal among citizens who had voted for direct implementation in Block 1's final round ($V_{i,1,10} = \mathsf{A}$, $A_{i1} = 1$). Reversal means voting for group approval in Block 2's fifth and final round ($V_{i,2,5} = \mathsf{D}$, $R_i = 1$). Comparing recovery ($T_g = 1$) with persistence ($T_g = 0$) identified this difference. Only one potential outcome was observed per citizen. Assignment and inference operated at pool level.

For each leader i, t_{igbr} is the proposed allocation and ρ_{igbr} the proposed transfer. A transfer proposal is $M_{igbr} = \mathbb{1}\{\rho_{igbr} > 0\}$, with intensity $\rho_{igbr}/(5t_{igbr})$ when $t_{igbr} > 0$. Thus, $t = 10$ and $\rho = 5$ proposes moving 10% of the 50 collected points. Approval ballots q_{igbr}, proposal passage, contributions after rejection, and subsequent institutional ballots provide process evidence. Comparisons between the endogenously chosen rules are descriptive and do not isolate the effect of requiring approval.

Immediate reversal R_i^I recorded the first response to the structural package before Block-2 interaction (see Figure 5). The primary outcome R_i recorded Round 5 after four completed rounds. Secondary outcomes included automatic-implementation support averaged over Rounds 3–5 and realized group-level restoration. These measures captured short-run adaptation rather than long-run convergence.

Expected payoffs assessed whether participants understood how recovery changed the rules' relative performance, while expected transfers traced beliefs about leader discretion. After all paid choices, eleven process items measured perceived crisis severity and recovery, expected institutional performance, and views on approval, equal allocation, and transfers. A separate optional twelve-item background module recorded age group, gender, education, household financial strain, party voted for in Finland's 2023 parliamentary election, left–right placement, whether democracy is preferable to other forms of government, satisfaction with democracy in Finland, evaluation of a strong leader unconstrained by parliament and elections, external political efficacy, generalized trust, and willingness to take risks. These prespecified measures captured partisan and ideological trade-offs, distinguished principled democratic support from evaluations of democratic performance, and benchmarked cooperative and risk-related predispositions (Graham and Svolik 2020; Saikkonen and Christensen 2023; Claassen 2020; Dohmen et al. 2011; Glaeser et al. 2000). We collected them only after paid decisions. They supported descriptive heterogeneity analyses but did not independently identify a treatment mechanism. Primary precision adjustment used only prespecified characteristics fixed before treatment and Block-1 measures, not potentially treatment-responsive attitudes collected afterward.

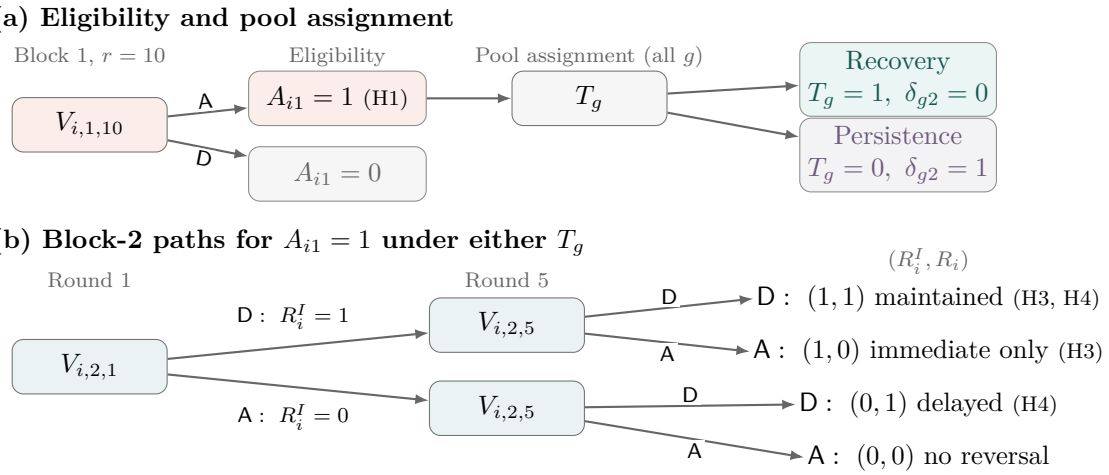


Figure 5: Decision tree for assignment and reversal. Each pool received one value of T_g. The primary analysis followed participants with $A_{i1} = 1$.

Notes: The displayed ballots are $V_{i,1,10}$ (final Block 1), $V_{i,2,1}$ (first Block 2), and $V_{i,2,5}$ (final Block 2). The ordered pair (R_i^I, R_i) records the two Block-2 outcomes, with $1 = \mathsf{D}$ and $0 = \mathsf{A}$. Thus, $(1,1)$ is reversal at both endpoints, $(1,0)$ is immediate reversal only, $(0,1)$ is delayed reversal, and $(0,0)$ is no reversal. Parentheses identify the corresponding hypotheses. H2 concerns transfers and is not shown.

5 Randomization, Sample, and Design Sensitivity

The study targeted 300 completed participants. Larger laboratory cohorts were preferred, with five-person sessions available when attendance was insufficient for rematching. The realized pool mix determined the number of pool-level randomization units, ranging from 20 if all pools contained fifteen participants to 60 if all contained five. The maximum payment of EUR 15 set the compensation ceiling for 300 completions at EUR 4,500.

Treatment was assigned only after Block 1. Separately for each pool size, successive pools entered four-pool *randomization sets* containing two recovery and two persistence assignments in random order (Figure 6). These sets continued across laboratory sessions. The software stored each pool's set and position, concealed unused assignments, and reused the recorded allocation after an interruption. An incomplete final set retained its randomly drawn prefix. Balance therefore held within each completed set, rather than necessarily within a session or the final sample.

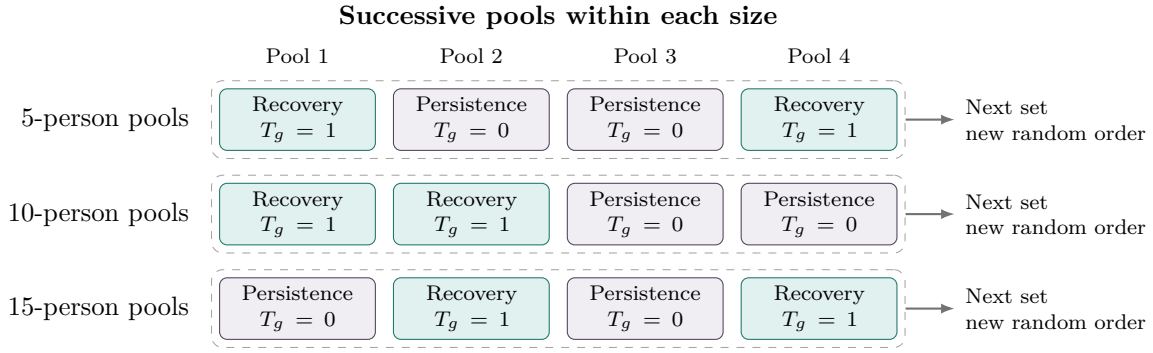


Figure 6: Treatment balancing separately by matching-pool size.

Notes: Orders are illustrative. Each cell represents one entire pool assigned after Block 1, not a game round. Each completed four-pool set contains two recovery and two persistence assignments. Each row continues across sessions independently of the others. A final incomplete set need not be balanced.

Compensation depended on one selected round per block, not the number of rounds played. Under the monetary benchmark above, each selected round yielded 44 expected points before leader selection, giving EUR $10 + 88/24 \approx 13.67$ per participant and approximately EUR 4,100 for 300 completions. This was a conditional illustration rather than a spending forecast. The EUR 4,500 ceiling covered all completed-participant payments but left no allowance for additional recruitment or operating costs.

The number of pools determined the amount of independent treatment information. A larger pool supplied more individual outcomes but still received one treatment assignment. Across 300 completions, Block 1 generated 600 leader proposals and Block 2 generated 300. These repeated observations revealed trajectories, but did not create additional assignment units. Five-person pools added units per participant while exposing the same citizens to a shared group history, which could increase similarity within pools.

We evaluated design sensitivity with a simulation that reproduced pool-level treatment assignment, the eligible subgroup, the Round-5 outcome, and the two-sided randomization test at the five-percent level. *Power* is the probability that this test detects a genuine recovery effect of an assumed size. It depends on the share eligible for the primary outcome, the reversal rate under persistence, and the *intrapool correlation* (ICC), which measures how similar outcomes are within a pool. A 20-percentage-point effect means a reversal rate 0.20 higher under recovery. We varied pool composition and allowed greater correlation in fixed five-person groups. The simulations assumed independence between pools, not between their members.

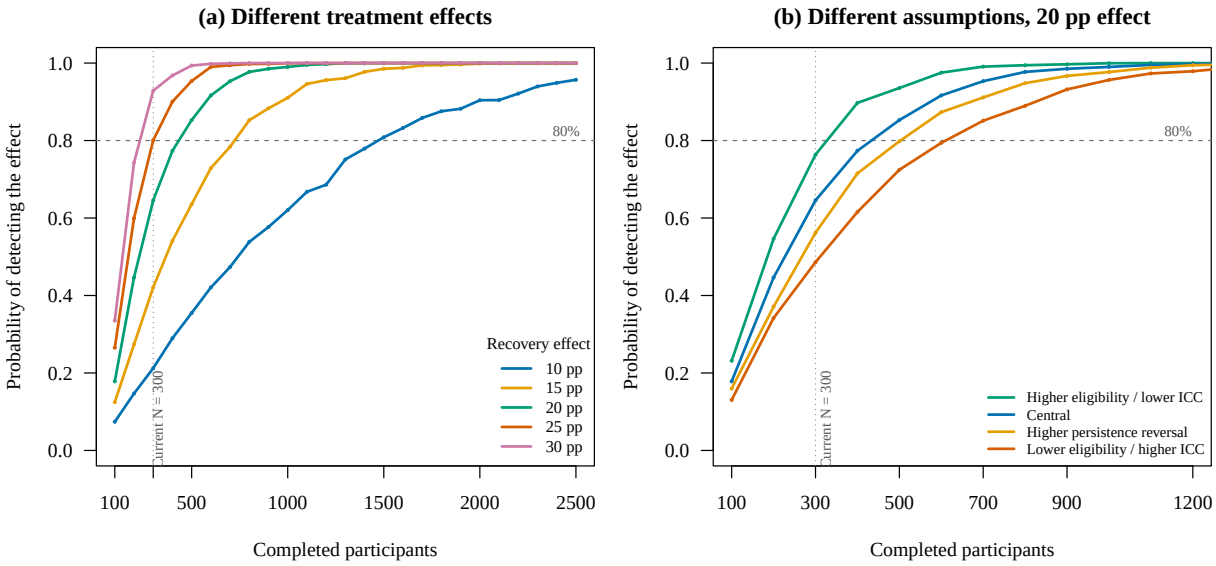


Figure 7: Simulated power as the completed sample increases

Notes: These reference curves use ten-person pools and four-pool randomization sets, retaining any incomplete set. Panel (a) holds eligibility at 0.50, persistence reversal at 0.20, and intrapool correlation at 0.05, then varies the recovery effect. Panel (b) fixes that effect at 20 percentage points and varies the assumptions. Each additional 20 participants adds two pools. Points use 2,500 simulated experiments and 999 reallocations, with monotone summary lines. Horizontal and vertical lines mark 80% power and 300 completions. Code is in `analysis/power_simulation.R`.

Table 3: Conditional power with 300 completions and mixed pool sizes

Attendance scenario	Pool counts (5/10/15 persons)	Five-person ICC	Power by effect size		
			20 pp	25 pp	30 pp
Ten-person pools	0/30/0	—	64.8%	81.0%	91.3%
20% in five-person pools	12/24/0	0.05	64.0%	81.1%	92.4%
20%, greater similarity	12/24/0	0.20	63.3%	80.5%	90.8%
Mixed sizes, greater similarity	12/18/4	0.20	61.7%	79.3%	90.3%
Fifteen-person pools	0/0/20	—	57.5%	74.2%	87.3%

Pool counts list five-, ten-, and fifteen-person pools. ICC is 0.05 in larger pools. All rows assume 50% eligibility and 20% persistence reversal. Each estimate uses 5,000 experiments and 1,999 reallocations. The all-fifteen row is an attendance bound, not a recruitment target. Code is in `analysis/mixed_pool_power.R`.

The simulations supported five-person sessions as an attendance fallback rather than a guaranteed power gain. With 20% of participants in five-person pools, power remained near 64% for a 20-point effect and 81% for 25 points, even allowing substantially greater fixed-group correlation. A mix including fifteen-person pools produced slightly lower power. Effects of approximately 25 percentage points reached 80% power under these assumptions, while smaller effects remained less reliably detectable. In the ten-person reference design, a 20-point effect required approximately 440 completions, beyond the compensation budget. Further checks allowed lower eligibility and weaker recovery effects in fixed groups. Those changes reduced power, so the attendance mix and behavioral parameters remained planning uncertainties rather than assured gains.

The simulation conditioned on the treatment effect at the fifth ballot. Five Block-2 rounds supplied four rounds of experience before that choice, but did not establish behavioral convergence. In

illustrative scenarios where each completed round closed 20%, 50%, or 80% of the remaining distance from the first ballot to an assumed later limit, the fifth ballot captured 59%, 94%, or nearly 100% of that additional adjustment. The first ballot could already reflect an immediate treatment response. These were sensitivity assumptions, not estimates from participants. Slower adjustment could produce a smaller Round-5 effect and lower power. The checks in `analysis/round_horizon_checks.R` kept this uncertainty separate from the exact monetary benchmark.

The primary randomization test reproduced the recorded size-specific randomization sets, including incomplete sets. Regression estimates clustered uncertainty by pool, adjusted for pool-size indicators, and used small-sample inference. Equal-pool weighting and, when estimable, an analysis excluding five-person pools checked sensitivity to the realized mix. Because attendance determined pool size, differences between fixed and rematched groups were descriptive rather than causal effects of matching.

Strategic decision screens showed a 90-second countdown and reminded participants to choose an answer and click Next. At zero, the reminder remained visible and the page stayed open. The software recorded response time and flagged delays without generating a ballot or allocation. The researcher or a trained research assistant monitored progress and helped delayed participants complete the current decision.

6 Implementation, Ethics, and Project Timeline

Adults registered for UTU Choice Lab through UTU RPP/TOKA, which managed recruitment and attendance separately from the experiment. Before practice, oTree presented study-specific information and recorded consent. Participants learned that the detailed research question would be explained after the paid task because advance disclosure could influence the institutional ballot. All task rules, payments, data processing, foreseeable risks, and the right to stop were disclosed beforehand. The debrief explained the full research purpose after the final questionnaires and any optional feedback.

The researcher or a trained research assistant fixed a consenting roster of 5, 10, 15, or 20 before paid play. They formed the matching pools in Table 1 and monitored synchronized progress. No participant entered or changed pools after paid play began. Technical testing verified fixed-group and rematching sessions, cross-session allocation, both conditions, wait pages, response-time reminders, browser recovery, exports, and payments. In the first ten-person session, an optional six-question pre-test assessed instruction clarity, the distinction between methods, payment clarity, screen difficulties, suggested improvements, and participants' understanding of the study's purpose. It followed all outcome and background questions and preceded debriefing. A separate form pooled feedback without study codes, response timestamps, or links to experimental or payment records. This feedback option was disabled for subsequent sessions. Any substantive protocol changes arising from the pre-test required review before further collection and explicit treatment in the analysis plan.

Live oTree data were hosted on University of Turku IT servers in Finland. The public interface did not collect research data, and oTree used no artificial-intelligence functions. UTU RPP/TOKA received no oTree codes, experimental responses, or payment information. The research team made immediate payments using a temporary code-and-amount list. Any named receipt contained no oTree code. The temporary list was destroyed after payment reconciliation and no later than 30 November 2026. Anonymization also required removing or coarsening identifying combinations and a documented disclosure-risk review. Until that review confirmed anonymization, coded data remained access-restricted to Janne Tukiainen, Hector Bahamonde, and authorized personnel.

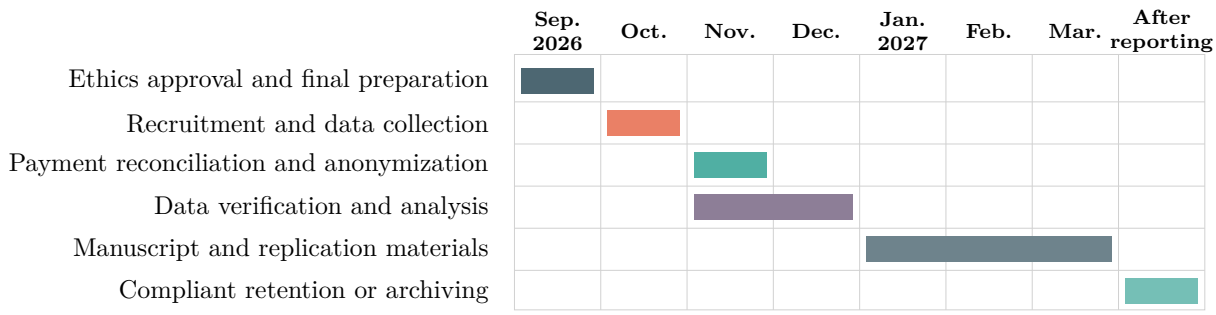


Figure 8: Study timeline.

Notes: Data collection took place from 1–31 October 2026 at UTU Choice Lab. Payment reconciliation and destruction of the temporary code-and-amount list were completed no later than 30 November 2026. Reporting, disclosure review, and retention or archiving followed the study's data-management plan.

The study produced academic articles, presentations, and replication materials. Consistent with University open-science policy, documentation, code, and data that could be shared safely were made as open as possible and as restricted as necessary.

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